

The Trotter-Error Gram Matrix

A reformulation of the open-system DMPF coefficient problem

Working note — open quantum system generalisation of dynamic MPF

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Abstract

The DMPF coefficients for open systems were being obtained by combining three separately computed objects — the Gram matrix M , the overlap vector $\vec{L}$ and the purity P — each of order 0.57 , whose combination is of order 10^{-4} . This is a catastrophic cancellation. We show the cost function can be rewritten exactly so that the small quantity, the *Trotter-error Gram matrix* N , is computed directly and the large one is never formed. Measured at $n = 4$ against an exact reference: the assembly step of the old route amplifies truncation error by $\approx 550\times$ while the new route de-amplifies by $\approx 5.5\times$, a ratio of 3018. The new route is $20\text{--}30\times$ more accurate at every bond dimension and never returns a negative squared norm, which the old route does at five of eight tested bond dimensions. **It does not reduce bond dimension and does not restore the closed-system near-identity cancellation;** Φ is in fact higher-rank than $\mathbb{F}$. The contribution is numerical conditioning and correctness, not asymptotic efficiency.

1 Summary: what this helps with, and what it does not

SOLVED — demonstrated numerically

Negative E_{mpf} (§7 of findings_thematic)	N is a Gram matrix of error vectors, hence PSD by construction. Measured $\lambda_{\min}$: M -route negative at 5 of 8 bond dimensions, worst -1.46×10^{-2} against a true $+2.81 \times 10^{-5}$ ($520\times$, wrong sign); N -route negative at 2, worst -5.3×10^{-6} (16% perturbation).
$E_{k_3} < E_{k_8}$ inversion (§8)	$E_{k_j} = N_{jj}$, a diagonal Gram entry. Correctly ordered and positive at every bond dimension tested.
Coefficients unstable in k_{ref} (§8)	$\vec{L}$ and P are eliminated; only k_0 remains.
MPS-vs-MPO method mismatch (§7)	P removed from the pipeline entirely. <i>Note:</i> P was never inaccurate — a Liouville MPS has Schmidt rank $\leq 4^{\min(l, n-l)} = 16$, below every <code>maxdim</code> tested, so <code>reference_purity</code> was <i>exact</i> . The failure was P exact while $\vec{L}$ was truncated, which makes the gauge cancellation structurally unavailable.
Accuracy per unit bond dimension	$20\text{--}30\times$ better at every χ . At $\chi = 64$: $\max \delta c = 3.0 \times 10^{-3}$ versus 9.1×10^{-2} .

NOT SOLVED — measured and negative

Bond dimension of the tracked object	χ_Φ saturates the $16^{\min(l,n-l)}$ ceiling at the <i>first</i> Trotter step, whereas $\chi_\mathbb{G}$ takes ~ 13 steps to climb $111 \rightarrow 256$. Φ is <i>higher</i> -rank than $\mathbb{F}$: effective rank at 10^{-3} retained weight is 193 for Φ versus 44 for $\mathbb{F}$. Mechanism: $\Delta_j = SS_j - \mathbb{E}$ is a difference of propagators, and differences generically carry higher Schmidt rank than either operand — the same cancellation that makes $\ \Phi\ $ small makes χ_Φ large.
The closed-system efficiency claim	Not restored. $SS^\dagger SS \neq \mathbb{K}$ for $\gamma > 0$ stands. The interleaved left/right schedule here serves only to avoid forming a one-sided full-time propagator ($\chi \sim e^{v_1 t}$), not to keep the running object near identity.
Wall-clock cost	<i>Worse</i> . At $n = 4$, $\chi = 256$: 770s versus ~ 80 s. Four tracked objects instead of one, plus an MPO addition per step. The driver is $\mathbb{G}$, not δ .
E_{mpf} itself	Unresolved in <i>both</i> routes. N is near rank-1 (all $\delta\rho_j$ are Trotter errors of the same dynamics, hence nearly parallel), so $\lambda_{\min}$ is tiny. $\vec{c}$ needs only the <i>eigenvector</i> direction and converges; E_{mpf} needs the <i>eigenvalue</i> magnitude and does not.
Monotonic convergence	Neither route converges monotonically in maxdim . Both show a pronounced bump at $\chi = 128$. Unexplained.

2 Diagnosis: why a reformulation was needed

2.1 The magnitudes

At $n = 4$, $\gamma = 0.05$, $t = 3$, $k = \{3, 8\}$, order 2, with the exact ($\chi = 256$, untruncated) reference:

$$M_{ij}, L_j, P \approx 0.57, \quad E_{k_3} = 1.33 \times 10^{-2}, \quad E_{k_8} = 2.50 \times 10^{-4}, \quad E_{\text{mpf}} = 3.76 \times 10^{-5}.$$

Inputs of order 0.57; the answer four orders below. The cost function $C(\vec{c}) = P + \vec{c}^T M \vec{c} - 2\vec{L}^T \vec{c}$ subtracts near-equal quantities.

2.2 The gauge structure

On the constraint set $\vec{1}^T \vec{c} = 1$, the perturbation

$$M \rightarrow M + \vec{u} \vec{1}^T + \vec{1} \vec{u}^T + a \vec{1} \vec{1}^T, \quad \vec{L} \rightarrow \vec{L} + \vec{u} + a \vec{1}, \quad P \rightarrow P + a$$

leaves C *exactly* unchanged for any $\vec{u}, a$. So $(M, \vec{L}, P)$ carries $r + 1$ redundant directions: $r(r + 1)/2 + r + 1$ numbers encode only $r(r + 1)/2$. The invariant content is exactly $N_{ij} = M_{ij} - L_i - L_j + P$. The gauge directions carry ~ 0.57 ; the invariant content carries 10^{-4} to 10^{-2} .

2.3 Measured: the assembly amplifies, it does not cancel

The `gauge_diagnostic` runs decompose the truncation error into gauge and residual parts:

case	$\ \delta M\ $	$\ \delta \vec{L}\ $	$ \delta P $	$\ \delta N\ $	amplification	resid/gauge
$n=4, 32 \rightarrow 64$ (ord. 1)	7.2×10^{-2}	7.5×10^{-2}	0	1.60×10^{-1}	1.54	0.91
$n=5, 64 \rightarrow 256$ (ord. 1)	1.7×10^{-3}	1.8×10^{-3}	0	5.0×10^{-3}	2.04	1.46
$n=5, 128 \rightarrow 256$ (ord. 1)	3.8×10^{-3}	1.1×10^{-2}	0	2.7×10^{-2}	2.40	0.92
$n=4, 32 \rightarrow 64$ (ord. 2)	2.6×10^{-3}	4.2×10^{-3}	0	9.6×10^{-3}	1.95	1.77
$n=4, 64 \rightarrow 256$ (ord. 2)	5.5×10^{-3}	2.2×10^{-3}	0	1.13×10^{-2}	1.90	0.67

The combination amplifies input error by a stable factor ≈ 2 . With $\delta P = 0$ identically (purity is exact), $\delta N_{ij} = \delta M_{ij} - \delta L_i - \delta L_j$: each $\delta \vec{L}$ enters twice with the same sign. Check the last row: $5.5 \times 10^{-3} + 2(2.2 \times 10^{-3}) = 9.9 \times 10^{-3}$ against a measured 1.13×10^{-2} — near-perfect constructive addition. *There is no cancellation to rely on.* Note also that the $n = 4, 64 \rightarrow 256$ row has resid/gauge = 0.67: a majority of that error *was* gauge, and it still destroyed the answer.

2.4 $\mathbb{F}$ and $\mathbb{G}$ are the same object

	$\ \cdot\ _F$	σ_1	σ_1^2/tot	eff. rank @10 ⁻³	σ_2/σ_1
$\mathbb{F} = SS^\dagger SS$	9.3986	9.3317	98.58%	44	0.0499
$\mathbb{G} = \mathbb{E}^\dagger \mathbb{E}$	9.3972	9.3340	98.66%	40	0.0474
$\Xi = \Delta^\dagger \mathbb{E}$	0.1796	0.0599	11.1%	129	0.981
$\Phi = \Delta^\dagger \Delta$	0.0061	0.0036	35.4%	193	0.380

$$\frac{\sum_i (\sigma_i^{\mathbb{F}} - \sigma_i^{\mathbb{G}})^2}{\|\mathbb{F}\|_F^2} = 2.24 \times 10^{-5}, \quad \frac{\|\Phi\|_F^2}{\|\mathbb{F}\|_F^2} = 4.2 \times 10^{-7}.$$

The part of $\mathbb{F}$ that determines the coefficients is four parts in ten million of its Frobenius weight. **Deck slides 22, 23 and 27 were characterising $\mathbb{E}^\dagger \mathbb{E}$.** Slide 27’s conclusion sharpens from “computing $\mathbb{F}$ is unfeasible” to “ $\mathbb{F}$ is the wrong object: 99.99996% of its weight cancels.”

3 The reformulation

3.1 Derivation

Because $\sum_j c_j = 1$ we may write $\rho(t) = (\sum_j c_j) \rho(t)$, hence

$$\mu(t) - \rho(t) = \sum_j c_j [\rho_{k_j}(t) - \rho(t)] = \sum_j c_j \delta \rho_j.$$

The MPF residual is a combination of *errors*, not of *states*. Therefore

$$C(\vec{c}) = \vec{c}^T N \vec{c}, \quad N_{ij} = \text{Tr}[\delta \rho_i \delta \rho_j] = \langle \langle \rho_0 | \Delta_i^\dagger \Delta_j | \rho_0 \rangle \rangle, \quad \Delta_j = SS_j - \mathbb{E}$$

a pure quadratic form: no linear term, no constant. Then

$$\vec{c} = \frac{N^{-1} \vec{1}}{\vec{1}^T N^{-1} \vec{1}}, \quad E_{\text{mpf}} = \frac{1}{\vec{1}^T N^{-1} \vec{1}}, \quad E_{k_j} = N_{jj}.$$

Substituting $SS_j = \mathbb{E} + \Delta_j$ shows what is removed: $\mathbb{F}_{ij} = \mathbb{G} + \Xi_i + \Psi_j + \Phi_{ij}$ with $\mathbb{G}$ of order 1 and i, j -independent (it cancels), Ξ, Ψ first order, and only Φ surviving.

3.2 Assumptions

A1 Hermiticity, so all scalars are real. **A2** $\text{Tr} \rho_{k_j} = 1$: holds because $\text{Tr}(\mathcal{L}[\rho]) = 0$ identically, so $e^{s\mathcal{L}}$ is trace-preserving for *every* real s — this survives the negative substep of order 4 (the §5.3 caveat concerns positivity only). **A3** adjoint, not inverse. **A4** $\mathbb{E} = S(t/k_0)^{k_0}$ is finite, so N measures error relative to that reference. **A5** k_0 divisible by every k_j . **A6** time-independent generator, so factors may be appended on either side.

Limitation L1. The N and $(M, \vec{L}, P)$ forms agree *only* on $\vec{1}^T \vec{c} = 1$. Relaxing the constraint breaks the equivalence.

3.3 Algorithm: the four-object recursion

With $A_j = S(t/k_j)$, $\mathcal{B}_j = S(t/k_0)^{k_0/k_j}$ (the reference over the *same* duration) and $\delta_j = A_j - \mathcal{B}_j$, track at independent clocks (t_a, t_b) :

$$\mathbb{G} = \mathbb{E}(t_a)^\dagger \mathbb{E}(t_b), \quad \Xi_i = \Delta_i(t_a)^\dagger \mathbb{E}(t_b), \quad \Psi_j = \mathbb{E}(t_a)^\dagger \Delta_j(t_b), \quad \Phi_{ij} = \Delta_i(t_a)^\dagger \Delta_j(t_b).$$

	advance right clock (one A_j)	advance left clock (one A_i)
$\mathbb{G}$	$\mathbb{G} \mathcal{B}_j$	$\mathcal{B}_i^\dagger \mathbb{G}$
Ξ_i	$\Xi_i \mathcal{B}_j$	$A_i^\dagger \Xi_i + \delta_i^\dagger \mathbb{G}$
Ψ_j	$\Psi_j A_j + \mathbb{G} \delta_j$	$\mathcal{B}_i^\dagger \Psi_j$
Φ_{ij}	$\Phi_{ij} A_j + \Xi_i \delta_j$	$A_i^\dagger \Phi_{ij} + \delta_i^\dagger \Psi_j$

Initial: $\mathbb{G} = \mathbb{K}$, $\Xi = \Psi = \Phi = 0$. Schedule: advance whichever clock is behind — identical to the loop in `build_open.F`. Consistency check: on a right advance, $(\mathbb{G} + \Xi)(\mathcal{B}_j + \delta_j) + (\Psi + \Phi)A_j = \mathbb{F}A_j$, since $\mathcal{B}_j + \delta_j = A_j$. **This identity is load-bearing: every update rule was derived from it, and it must hold numerically.**

Why left *and* right multiplication. Each object is a sandwich, so advancing t_a appends left and t_b appends right. Running all right steps first would leave $\mathbb{G} = \mathbb{E}(t_b)$ — a bare full-time propagator with $\chi \sim ge^{v_1 t}$ (Appendix A of the closed-system paper). Interleaving keeps the objects *balanced*. This is the only job the ordering does here; in the closed system it additionally forces $\chi = 1$ via $U^\dagger U = \mathbb{K}$, and that mechanism is gone for $\gamma > 0$.

4 Results

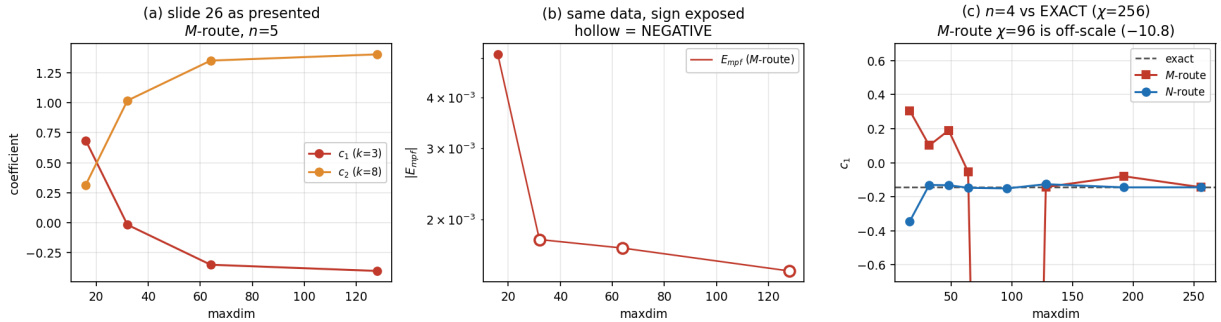


Figure 1: Deck slide 26, as presented and re-examined. (a) The original $n = 5$ M-route coefficients. (b) The same run's E_{mpf} with the sign exposed: *three of the four points are negative*, which the absolute-value axis concealed. E_{mpf} is a squared norm. (c) $n = 4$ against the exact untruncated reference: the M-route wanders (its $\chi = 96$ point, $c_1 = -10.8$, is off-scale) while the N-route sits on the exact value from $\chi = 32$.

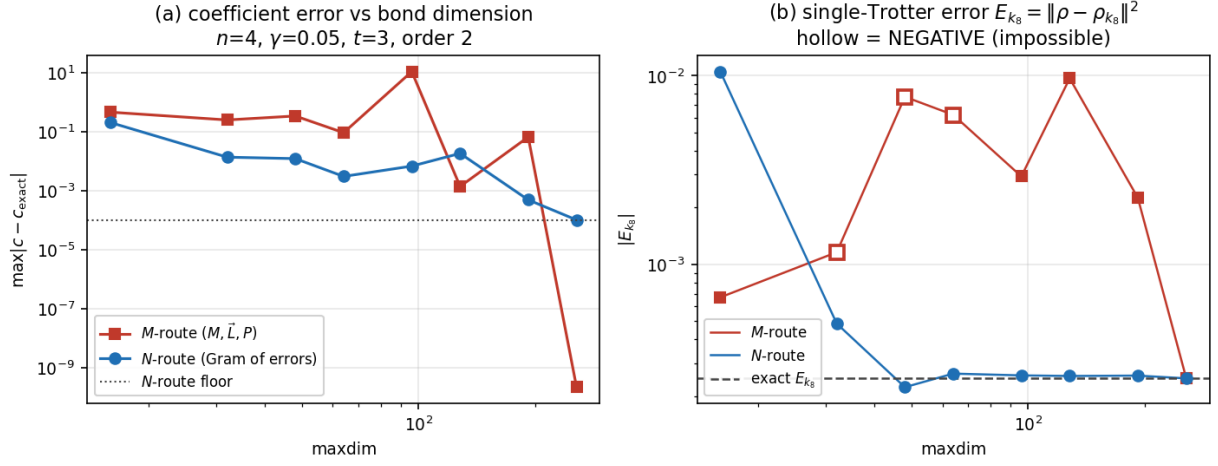


Figure 2: Accuracy against exact ground truth, $n = 4$. (a) Coefficient error. The M -route is non-monotonic and never below 10^{-3} except at the exact point; the N -route is bounded in $[10^{-4}, 2 \times 10^{-2}]$ for all $\chi \geq 32$. (b) The single-Trotter error E_{k8} : hollow markers are *negative* values of a squared norm. The M -route is negative at three bond dimensions and wrong by 800–3800 % elsewhere; the N -route is positive everywhere and within 8 % from $\chi = 48$ on.

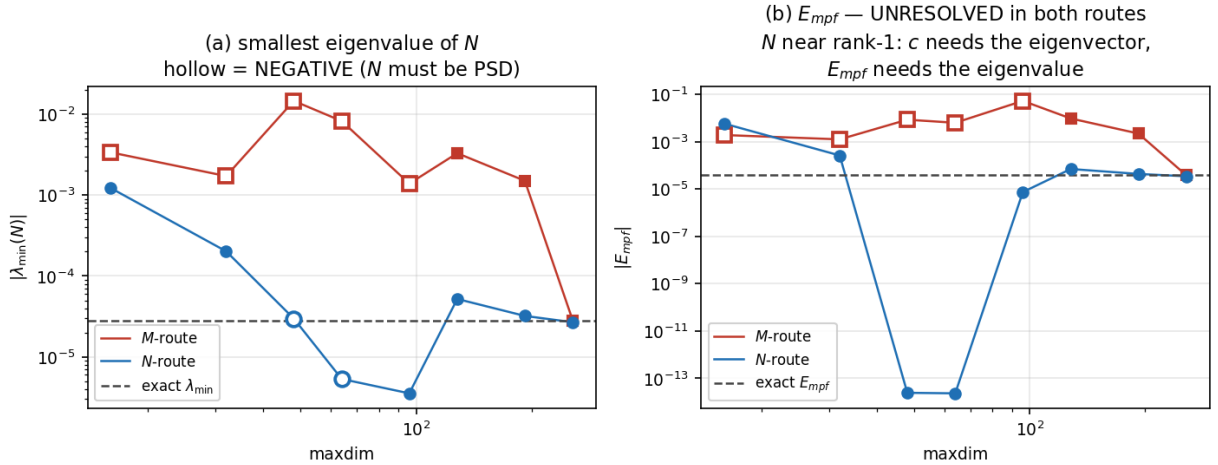


Figure 3: (a) $\lambda_{\min}(N)$, which must be positive. (b) E_{mpf} is unresolved in *both* routes — a genuine limitation, not a defect of either. N is near rank-1, so $\vec{c}$ (eigenvector direction) converges while E_{mpf} (eigenvalue magnitude) does not.

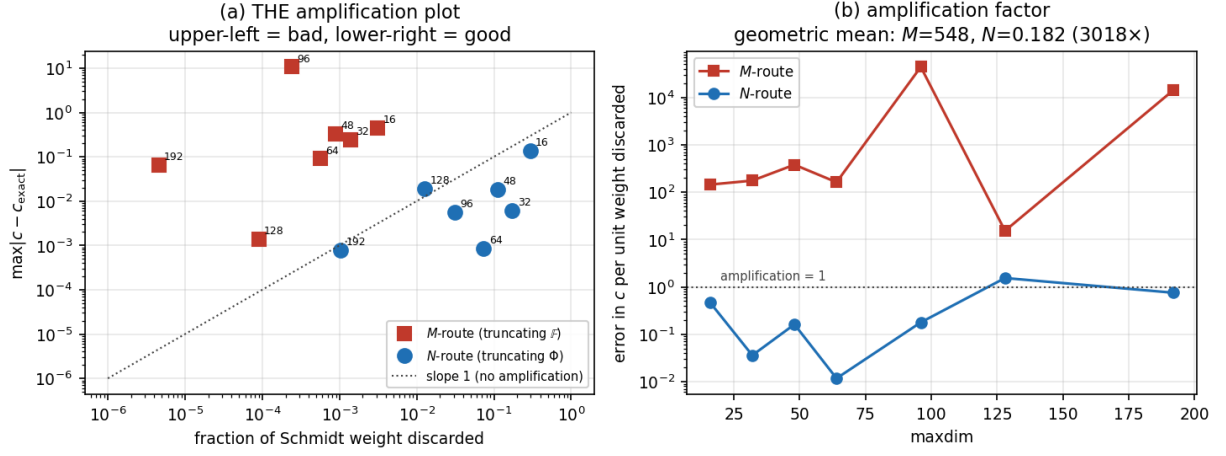


Figure 4: The central quantitative result. (a) Schmidt weight discarded by truncation versus the resulting error in $\vec{c}$. The M -route discards $\sim 10^{-4}$ of $\mathbb{F}$ and loses the answer; the N -route discards up to 29% of Φ and keeps it. (b) Amplification, i.e. error in $\vec{c}$ per unit weight discarded: geometric mean 548 (M) versus 0.182 (N), a ratio of 3018. Φ **does not compress** — **it does not need to**.

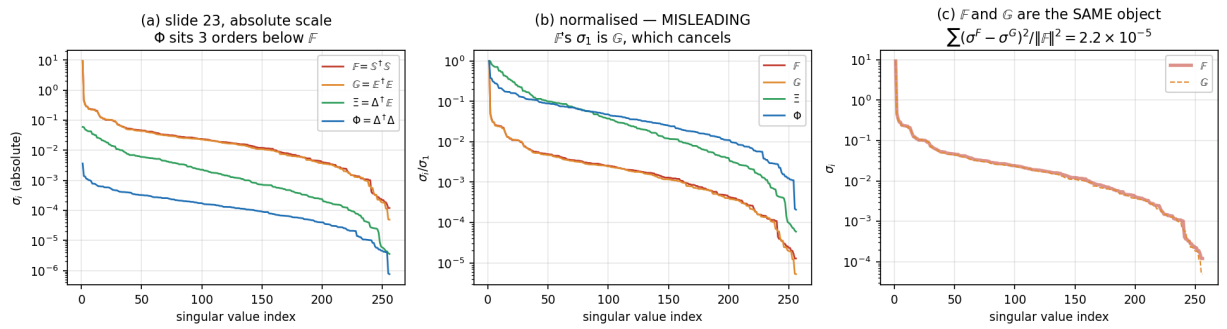


Figure 5: Deck slide 23 revisited, $n = 4$, $k = 8$, untruncated. (a) Absolute singular values. (b) Normalised — *misleading*, because $\mathbb{F}$'s σ_1 is the $\mathbb{G}$ piece that cancels out of the answer, making its tail look negligible when that tail is the entire signal. (c) $\mathbb{F}$ and $\mathbb{G}$ coincide to 2.2×10^{-5} : the bond-dimension study was measuring $\mathbb{E}^\dagger \mathbb{E}$.

5 Numerical pitfalls discovered (general, and costly)

Four rounds of debugging were spent on these. They apply anywhere in this project where two near-equal MPOs are subtracted — which is the premise of the N -route.

1. **maxdim governs the loss in MPO addition; cutoff does nothing.** Adding a large and a small MPO preserves the small term only if $\text{maxdim} \geq \chi(A) + \chi(B)$. Once rank truncation occurs, the retained directions represent the *sum* (dominated by the large term) and the small term's distinct directions are discarded. Measured: at $\text{maxdim} = 16$ the small term was lost with 95% relative error for cutoffs 10^{-12} , 10^{-14} , 10^{-16} and *exactly* 0 — identical results.
2. **+ on MPS/MPO defaults to `alg="densitymatrix"`**, which forms $\rho = MM^\dagger$ and diagonalises, costing half the available digits ($\sqrt{\varepsilon_{\text{mach}}} \approx 1.5 \times 10^{-8}$). Use `alg="directsum"` for differences of near-equal operators; it takes *no* cutoff/maxdim (passing them raises a `MethodError`).
3. **$\sqrt{\text{inner}(X, X)}$ cannot resolve a difference below $\sqrt{\varepsilon}\|A\|$.** It computes a squared quantity by contraction, so measuring $\|B + \delta - A\|$ requires cancelling terms of order $\|A\|^2$ down to $(\varepsilon\|A\|)^2$. Any claim of the form “verified to 10^{-15} ” about a difference of large MPOs is an artifact. This bit us twice in opposite directions: a density-matrix subtraction once reported 2.9×10^{-15} for a true value of 1.5×10^{-8} .
4. **maxdim must never exceed $16^{\min(l, n-l)}$** ($16/256/256/4096$ for $n = 3/4/5/6$). A loose maxdim forces enormous intermediate tensors before truncation. Also: pin BLAS threads in *both* Julia and the batch script.

6 Open issues

1. **A floor at $\max|\delta c| \approx 1.0 \times 10^{-4}$ at $\chi = 256$, where nothing is truncated.** Ruled out: `op_dag` (3×10^{-15}); reference pre-composition (8.5×10^{-7}); `cutoff` ($10^{-12} \rightarrow 10^{-15}$ moves it only $1.6 \rightarrow 0.97 \times 10^{-4}$); the δ construction (bounded at 10^{-8} by the measurement floor, giving $\lesssim 5 \times 10^{-6}$ after amplification by $\text{cond}(N) = 483$). Switching to `alg="directsum"` changed the sweep by less than 1%. **Cause unidentified; likely upstream in A or B .**
2. **Non-monotonic convergence in both routes**, with a pronounced bump at $\chi = 128$. Unexplained, and the kind of thing a referee will ask about.
3. **The reference is not converged.** $k_0 : 48 \rightarrow 96$ shifts E_{k_8} by +6.3%. The $n = 4$ “exact” point is exact in the sense of *untruncated*, not *physically correct*. Fine for an M -vs- N comparison (both use it), but $E_{k_8} = 2.5 \times 10^{-4}$ is not the physical value.
4. **The $k = 3$ branch is badly conditioned (L5): $\|\delta_3\|/\|A_3\| = 9.3 \times 10^{-2}$ versus 6.0×10^{-3} for $k = 8$,** because the $k = 3$ block spans $t/3 = 1.0$. Fix: subdivide the coarse step so every defect spans a short interval.
5. **Only $\gamma = 0.05$ has been tested.** Slide 22 swept $\gamma \in \{0.01, 0.05, 0.1, 0.2\}$. Whether the $3018\times$ amplification advantage survives at $\gamma = 0.2$ is untested and is the single most important open question.

7 Recommended next steps

1. **γ sweep** (highest value). Repeat the $n = 4$ head-to-head at $\gamma \in \{0.01, 0.1, 0.2\}$; plot $\max|\delta c|$ at fixed $\chi = 32$ versus γ for both routes. Direct successor to slide 22 and the most persuasive summary figure available.

2. **Subdivide the $k = 3$ coarse step** and re-measure N_{11} .
3. **Do not chase the 10^{-4} floor further.** It sits four orders below the error of the route it replaces, which additionally returns negative squared norms. No conclusion in this document depends on it.
4. **For an asymptotic efficiency claim,** neither of the two structural obstacles is addressed here: $\mathbb{G}$ saturates, and Φ saturates immediately because it is a difference. The candidates remain the KMS/detailed-balance inner product (under which $\mathbb{G} \rightarrow e^{2t\mathcal{L}_D}$ becomes a genuine contraction, bond dimension 1 for on-site dissipators — and for which the *dephasing* test is nearly free, since $\text{KMS} = \text{HS}$ for unital dissipators), the γ -expansion, or temporal/influence-matrix contraction.

8 Code

New: `trotter_error_gram.jl` (library), `gauge_diagnostic.jl`, `comparison_study.jl`, `spectra_study.jl`, `systematic_diagnostic.jl`, `identity_localise.jl`, `delta_microtest.jl`, `directsum_test.jl`, `delta_fix_verify.jl`, plus `submit_*.sh`.

Replaced: `open_gram_matrix`, `open_L_vector`, `reference_purity`, `dynamic_mpf_coefficients`, `open_dynamic_mpf_error`, `open_single_trotter_errors` — six functions collapse to `trotter_error_gram` + `trotter_error_coefficients`.

Reused unchanged: `liouville_space.jl`, `open_product_formula_generation.jl`, the `op_dag/left_multiply/right_multiply` helpers and two-clock schedule in `open_middle_out_contraction.jl`, `bond_dimension_tracking.jl`, `F_diagnostics.jl`.

Include chain: `include("F_diagnostics.jl")` pulls the whole stack. Including `open_middle_out_contraction.jl` directly leaves `vectorized_initial_state_mps` and `middle_bond_dim` undefined.